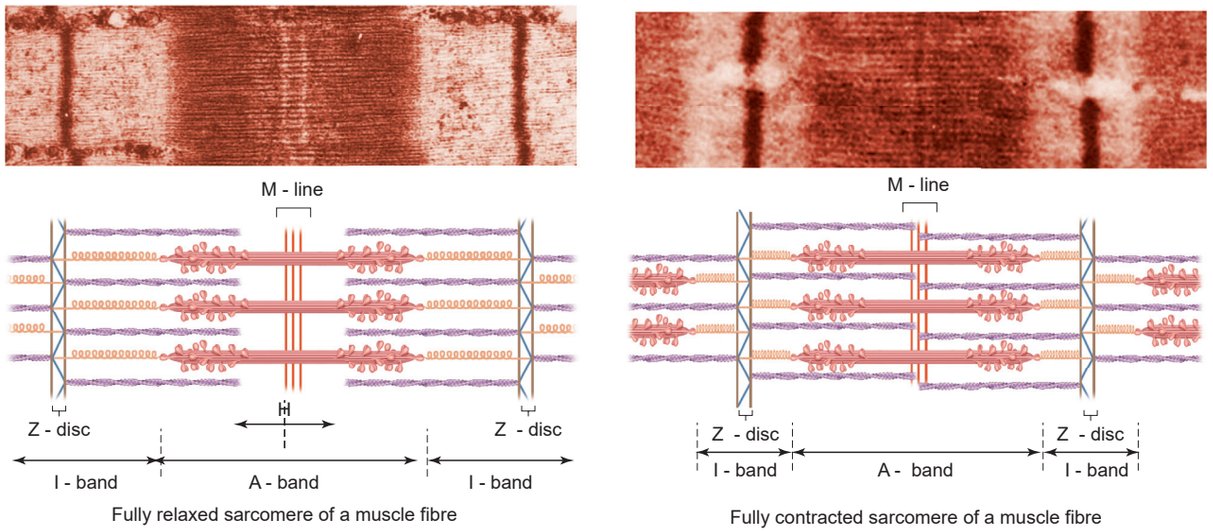


Figure 9.4 Sliding filament model of muscle contraction

Schematic Presentation of Muscle Contraction

Muscle contraction is initiated by the signal from CNS

↓

Release of acetylcholine at the neuromuscular junction

↓

Causes action potential in muscle fibre

↓

Triggers the release of calcium ions from sarcoplasmic reticulum

↓

Calcium ions combine with troponin and tropomyosin uncovers the binding sites on actin and initiates contraction

↓

Myosin binding sites on actin exposed. Myosin head binds to actin

↓

Myosin head executes power stroke

↓

Actin filament slides towards the centre of sarcomere (contraction)

↓

Signal from CNS stops; calcium ions are pumped back into the sarcoplasmic reticulum

↓

Tropomyosin masks the binding sites. Filaments pulled back to the original position (relaxation)

9.6. Types of skeletal muscle contraction

There are two primary types of muscle contractions. They are **isotonic contraction** and **isometric contraction**. The types of contractions depend on the changes in the length and tension of the muscle fibres at the time of its contraction.

Isotonic contraction (iso- same, ton- weight/resistance)

In isotonic contraction the length of the muscle changes but the tension remains constant. Here, the force produced is unchanged. Example: lifting dumb bells and weightlifting.

Isometric contraction (iso- same, metric-distance)

In isometric contraction the length of the muscle does not change but the tension of the muscle changes. Here, the force produced is changed. Example: pushing against a wall, holding a heavy bag.

Types of skeletal muscle fibres

The muscle fibres can be classified on the basis of their rate of shortening, either fast

TN_GOVT_XI_Bio Zoology_Ch09.indd 156 09-01-2020 5.30.53 PM